

Console Printing and Styling Standards

To maintain a professional command-line interface display, all user-facing warnings, rejections, and success metrics printed to the screen conform to a strict, homogeneous color-and-icon-coded layout schema powered by Rich. Symmetrically, this ensures consistent user interaction and instant readability:

Message Category	Rich Color Code	Standard Icon	Code Example / String Format	Typical Event / Action
System Warnings & Conflicts	[dark_orange]	!!	<code>[bold dark_orange]!![/bold dark_orange] [dark_orange]{page_id} attempted premature exit...[/dark_orange] [bold dark_orange]!![/bold dark_orange] [dark_orange]Conflict # {next_id} recorded – see {rel}: {summary}[/dark_orange]</code>	Premature exits, blocked registrations, and source contradiction recording alerts
Asset Skipping Alerts	[yellow]	⏏\ufe0e SKIP	<code>[yellow]⏏\ufe0e SKIP[/yellow] Discovery Pass for {page_id} skipped.</code>	Informational skip updates
Phase / Round Success	[green]	✓	<code>[green]✓[/green] Phase 1 Drafting Pass for {page_id} complete.</code>	Successful completions and verified passes
Verification Approvals	[green]	✓ Pass	<code>[green]✓ Pass[/green] on Round {idx}! Promoting...</code>	Round-level quality approvals
Temporary Image	[yellow]	x Fail	<code>[yellow]x Fail[/yellow] on Round {round_idx}. Violations...</code>	Round-level quality rejections

Message Category	Rich Color Code	Standard Icon	Code Example / String Format	Typical Event / Action
Verification Rejection				
Transient API Warnings	[magenta]	—	[magenta]Encountered a transient Gemini API/transport error inside {phase_name}.[/magenta]	Transient capacity or network overload warnings
API failure/Errors	[red]	—	[red]Phase 1 Drafting Pass failed: {error_brief}[/red] (Process-level failures) [red]Failed to caption region '{r.id}' due to unrecoverable error: {brief}[/red] (Single-thread errors)	Unexpected runtime, credential, parsing, or network errors, including thread-isolated failures
Final Image Verification Rejection	[dark_orange]	x Discarded	[dark_orange]x Discarded: [/dark_orange] Draft region '{label}'...	Permanent validation failures and rejections
Deep Agent Sub-processes	[yellow]	↳	[yellow]↳ Spawning Re-Crop Agent...[/yellow]	Dynamic re-crop agent spawning updates
Recovery operations	[blue]	↳	[blue]↳ Re-queued checklist item...[/blue] [blue]↳ Preserved: Page census...[/blue]	Census re-queuing and saving states to disk on unexpected aborts
Volatile Sweeps & Warm-ups	[dim]	—	[dim]Warming up models...[/dim] [dim]Sweeping temporary scratch buffers...[/dim]	Local-first database/model warm-ups and transient filesystem sweeps

Message Category	Rich Color Code	Standard Icon	Code Example / String Format	Typical Event / Action
Filesystem Paths	[cyan]	—	[cyan]{project_path}[/cyan]	Filesystem directories, file paths, and relative targets

High-visibility step milestones and initialization logs

Symmetrically, project workspace initializations, step transitions, and error catching caught by the CLI layer are rendered as high-visibility, full-line highlighted statements without leading icons, strictly conforming to the green success and red system failure formatting categories:

- **Project Switch (rag checkin):** text

```
[green]✓ Active project successfully checked in to '[cyan]{sanitized}[/cyan]' and settled successfully![/green]
[red]\\[FAIL] Failed to switch and settle active project workspace: {error_brief}[/red]
```

- **Workspace Initialization (rag init):** text

```
[green]✓ Environment audit passed successfully![/green]
[bold dark_orange]!![/bold dark_orange] [dark_orange][MISS] {dependency_name} -> {remediation_instructions}
[/dark_orange]
[bold dark_orange]!![/bold dark_orange] [dark_orange]Cannot proceed: install missing dependencies and re-run
`rag init`.[/dark_orange]
[red]\\[FAIL] Environment audit failed: {error_brief}[/red]
```

- **Ontology Definition (rag define-layers):** text

```
[green]✓ Ontology definition complete. Successfully seeded layers.yaml inside active project.[/green]
[red]\\[FAIL] Failed to define ontology: {error_brief}[/red]
```

- **Document Classification (rag documentation-layers):** text

```
[green]✓ Document layering complete. All documents classified and saved to inventory.[/green]
[bold dark_orange]!![/bold dark_orange] [dark_orange]Document layer proposal incomplete: {err_count} documents
were not classified due to API or parsing failures. Please re-run the command 'rag documentation-layers' to retry
```

those items.[/dark_orange]

[red]Failed to classify document layers: {error_brief}[red]

- **Pass 1 Discovery (rag macro-images):** text

[green]✓ Discovery Pass (Pass 1) complete. Bounding box coordinates successfully drafted on disk.[/green]

[bold dark_orange]!![/bold dark_orange] [dark_orange]Discovery Pass (Pass 1) complete with warnings: {count} macro-regions remained undrafted due to unrecoverable connection or API errors. Please run 'rag macro-images' again to retry those items.[/dark_orange]

- **Macro Verification (rag verify-drafts):** text

[green]✓ Verification round complete. All candidates promoted or successfully re-queued on disk.[/green]

[bold dark_orange]!![/bold dark_orange] [dark_orange]Verification round complete with warnings: {count} macro-regions remain unverified due to unrecoverable connection or API errors. Please run 'rag verify-drafts' again to retry those items.[/dark_orange]

- **Pass 2 Subdivision (rag micro-images):** text

[green]✓ Subdivision Pass (Pass 2) complete. All detail views successfully drafted on disk.[/green]

[bold dark_orange]!![/bold dark_orange] [dark_orange]Subdivision Pass (Pass 2) complete with warnings: {count} micro-items remained undrafted or unsubdivided due to unrecoverable connection or API errors. Please run 'rag micro-images' again to retry those items.[/dark_orange]

- **Micro Verification (rag verify-drafts --micro):** text

[green]✓ Verification round complete. All candidates promoted or successfully re-queued on disk.[/green]

[bold dark_orange]!![/bold dark_orange] [dark_orange]Verification round complete with warnings: {count} micro-regions remain unverified due to unrecoverable connection or API errors. Please run 'rag verify-drafts --micro' again to retry those items.[/dark_orange]

- **Asset Captioning (rag caption):** text

[green]✓ Asset captioning complete. Captions and vision prompts successfully appended to regions catalog.[/green]

[bold dark_orange]!![/bold dark_orange] [dark_orange]Asset captioning incomplete: {err_count} approved regions were not captioned due to API or parsing failures. Please re-run the command 'rag caption' to retry those items.[/dark_orange]

[red]\\[FAIL] Captioning run failed: {error_brief}[red]

- **Layer Assignment (rag asset-layers):** text

```
[green]✓ Asset layer assignment complete. Unified metadata catalog successfully synchronized on disk.[/green]
```

```
[bold dark_orange]!![/bold dark_orange] [dark_orange]Asset layer assignment incomplete: {err_count} assets were not classified due to API or parsing failures. Please re-run the command 'rag asset-layers' to retry those items.[/bold dark_orange]
```

```
[red]\\[FAIL] Layer assignment failed: {error_brief}[/red]
```

- **Search Index Build (rag build-index):** text

```
[green]✓ Vector index build complete. Multilingual semantic and full-text keyword indices compiled successfully![/green]
```

```
[red]\\[FAIL] Vector index build aborted: {error_brief}[/red]
```

```
[bold dark_orange]!![/bold dark_orange] [dark_orange]Vector index build incomplete: {error_brief}. Please re-run the command 'rag build-index' to compile your search indexes.[/bold dark_orange]
```

```
[red]Index build refused: not every inventory entry is HITL-approved.[/red]
```

```
[red]The indexes must match what `rag export` publishes.[/red]
```

```
[red] {err_count} document(s) unapproved – run `rag documentation-layers`[/red]
```

```
[red] {err_count} asset(s) unapproved – run `rag review`[/red]
```

```
[bold dark_orange]!![/bold dark_orange] [dark_orange]Index build skipped for document {doc_id} due to unrecoverable error.[/dark_orange]
```

```
[blue]↳ Rolled back partial index writes for {doc_id} from disk.[/blue]
```

```
[bold dark_orange]!![/bold dark_orange] [dark_orange]Index build skipped for asset {urn} due to unrecoverable error.[/dark_orange]
```

```
[blue]↳ Rolled back partial index writes for {slug} from disk.[/blue]
```

- **Deep Integrity Verification (rag verify):** text

```
[green]✓ Integrity verification checks passed successfully![/green]
```

```
[bold dark_orange]!![/bold dark_orange] [dark_orange]Some checks failed. Fix issues above before exporting.[/dark_orange]
```

```
[red]\\[FAIL] Verification run failed: {error_brief}[/red]
```

- **Distributable Package Exporting (rag export):** text

```
[green]✓ Machine catalog export complete. Finalized catalog package is completely compiled and ready for
```

distribution![/green]

[bold dark_orange]!![/bold dark_orange] [dark_orange]Pre-export verification is incomplete or failed.

Symmetrically, you must run 'rag verify' and resolve all checks successfully before exporting.[/bold dark_orange]

[red]\\[FAIL] Catalog export failed: {error_brief}[/red]

- **Asset Ingestion (rag ingest):** text

[green]✓ Ingestion complete (documents seeded).[/green]

[green]✓ Ingestion complete (URNs minted).[/green]

[red]\\[FAIL] Ingestion failed: {error_brief}[/red]

- **Workspace Reset (rag reset):** text

[green]✓ Reset complete.[/green]

[red]\\[FAIL] Reset failed: {error_brief}[/red]

- **Layer Re-Plotting:** text

[green]✓ Layer distribution plot successfully regenerated.[/green]